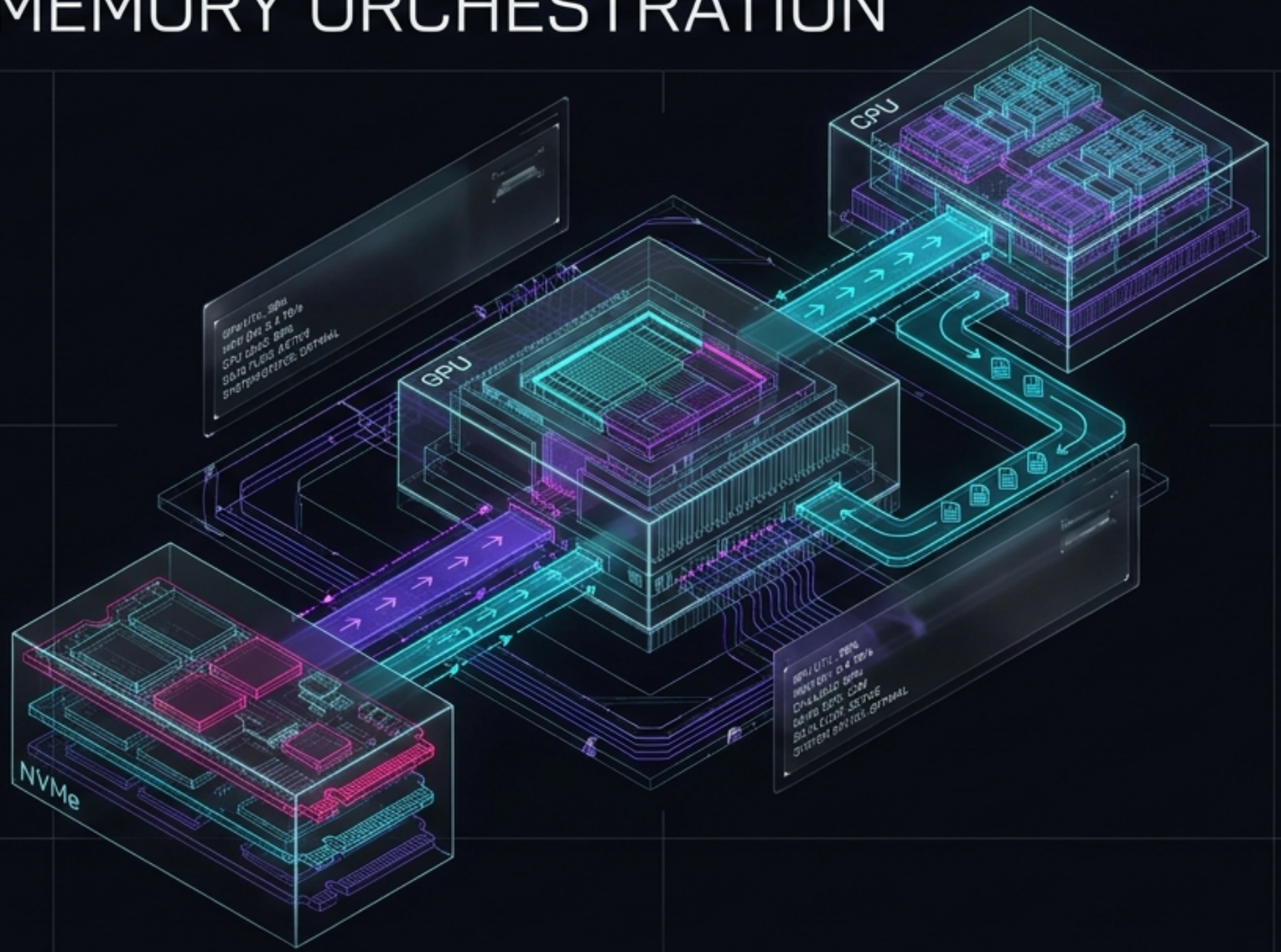
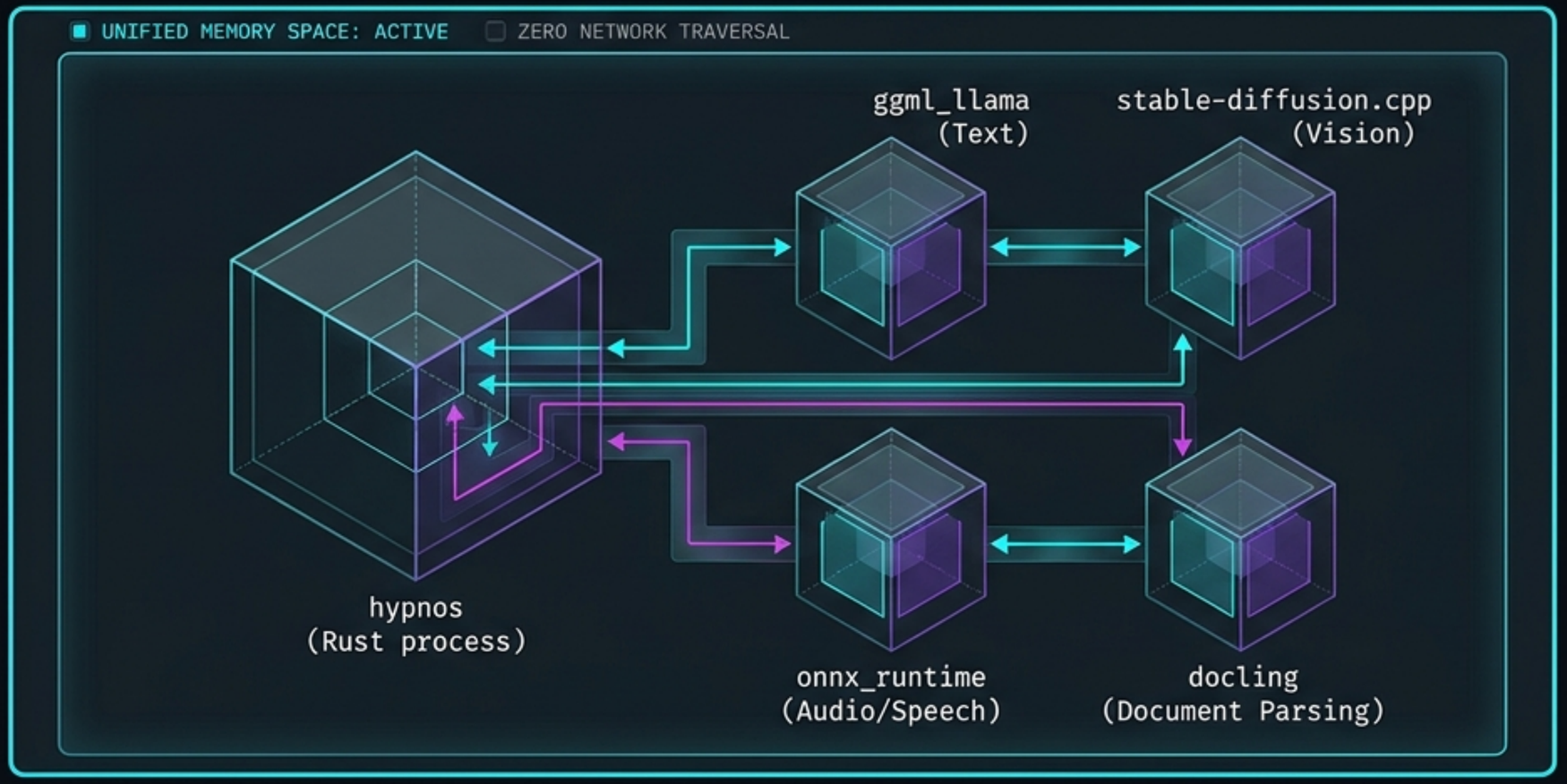
HYPNOS ARCHITECTURE: **MULTI-MODAL INFERENCE** & TIERED MEMORY ORCHESTRATION



Gemini Notebook

The In-Process Paradigm: Eradicating Network Hops



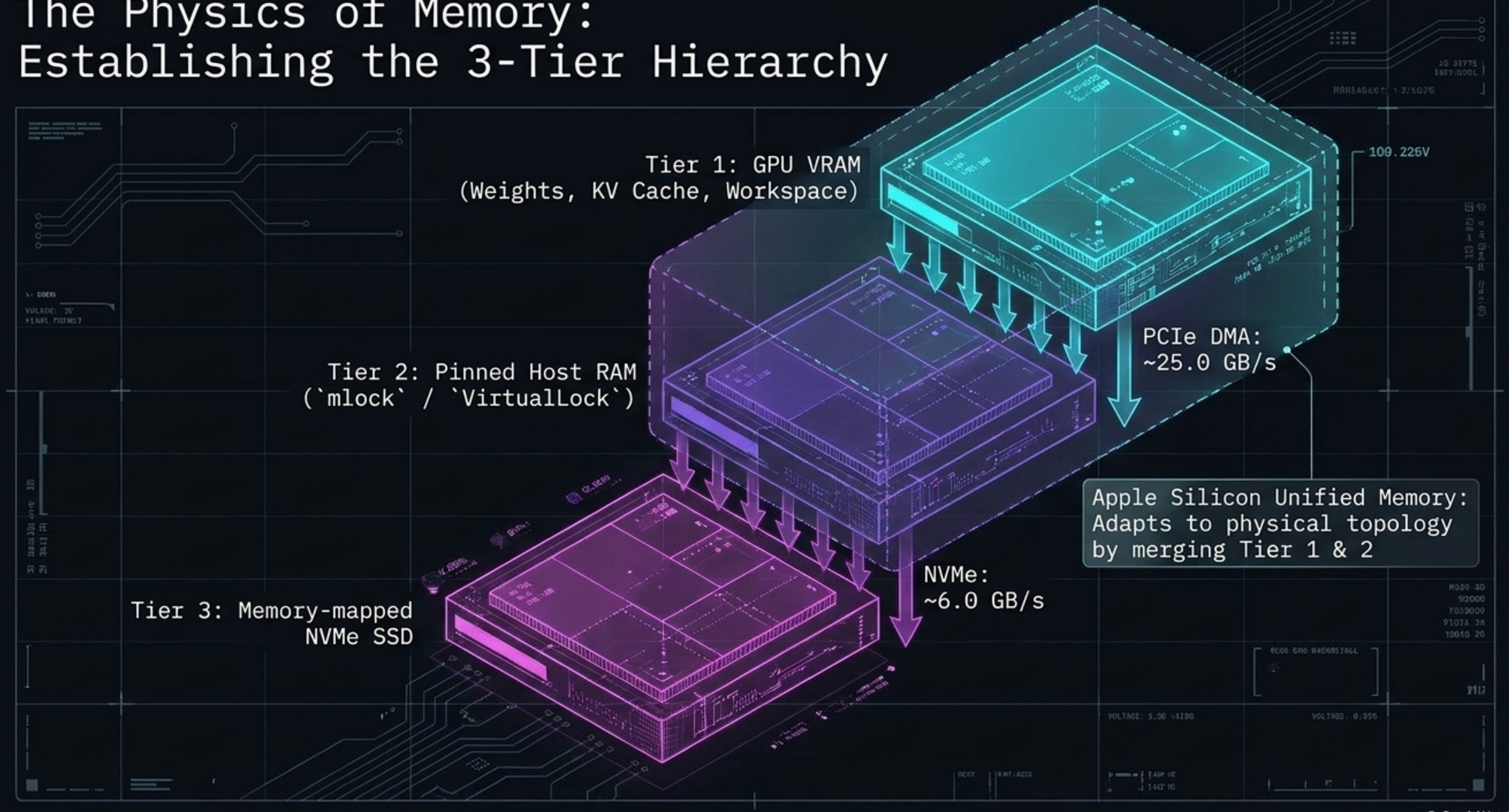
No external runtimes.

No network serialization overhead between modalities.

Zero-copy memory sharing across inference engines.

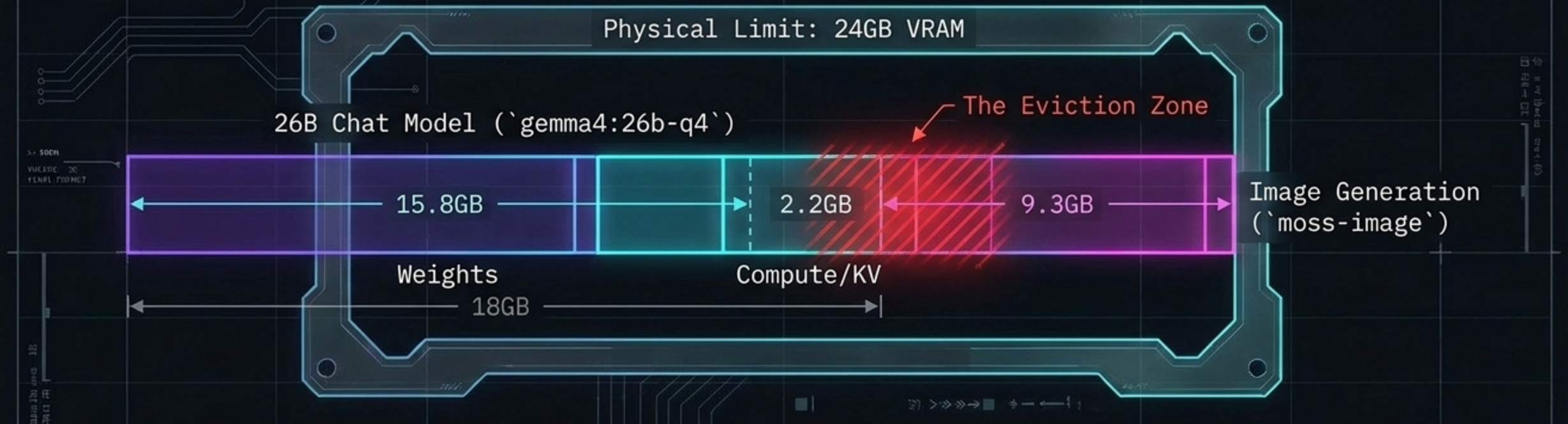
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The Physics of Memory: Establishing the 3-Tier Hierarchy



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Layer Placement & The Co-Residency Conflict

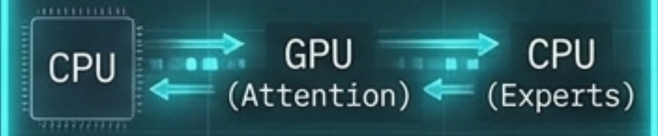


Goal	Reality
Maintain multi-modal responsiveness.	Models exceed physical VRAM, forcing continuous, high-latency swapping (eviction thrashing).

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Architectural Synthesis: Mixture-of-Experts (MoE) Offloading

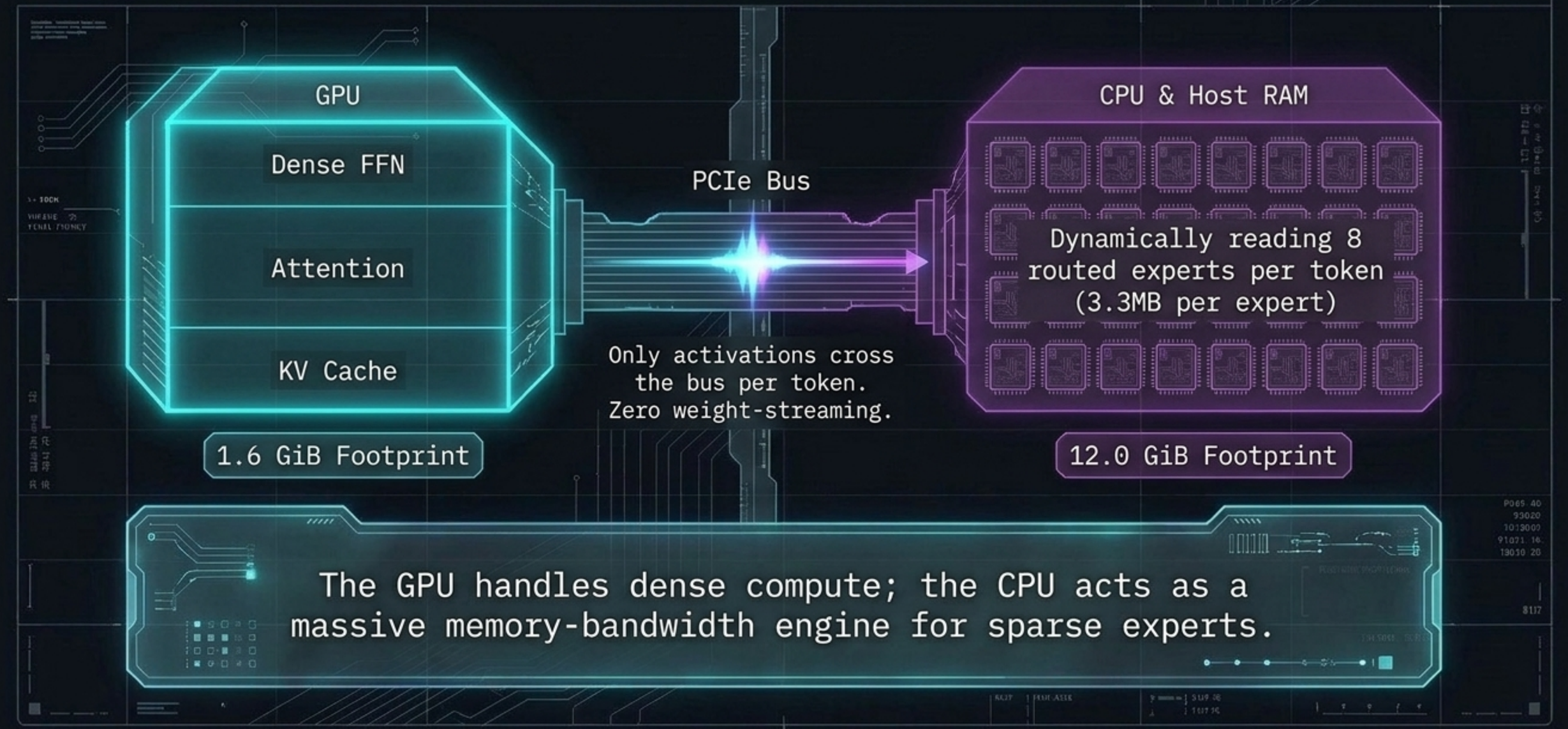
Serving `gemma4:26b` (88% expert weights)

Full GPU	Standard Layer Split	CPU Expert Offload (Hypnos)
13.6 GiB VRAM	Proportional VRAM	1.6 GiB VRAM (Attention/KV) + 12 GiB Pinned RAM (Experts)
Con: Blocks vision/audio models from co-residency.	Con: Decode is entirely bound by the slowest CPU layers in the chain.	

Breakthrough: Keep the massive experts in Host RAM. Execute them on the CPU. Reserve GPU for Attention.

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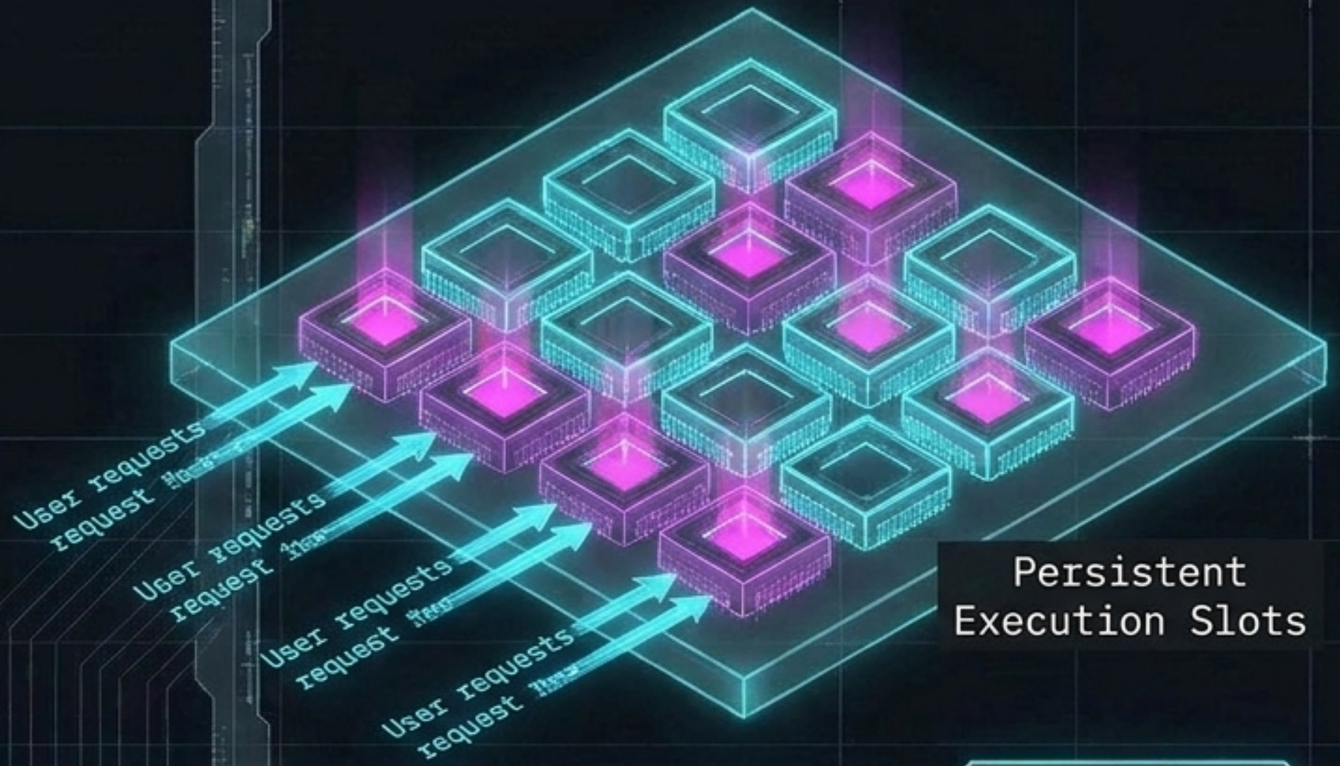
The Mechanics of CPU Expert Execution



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Rethinking Execution: Eradicating the Mutex Bottleneck

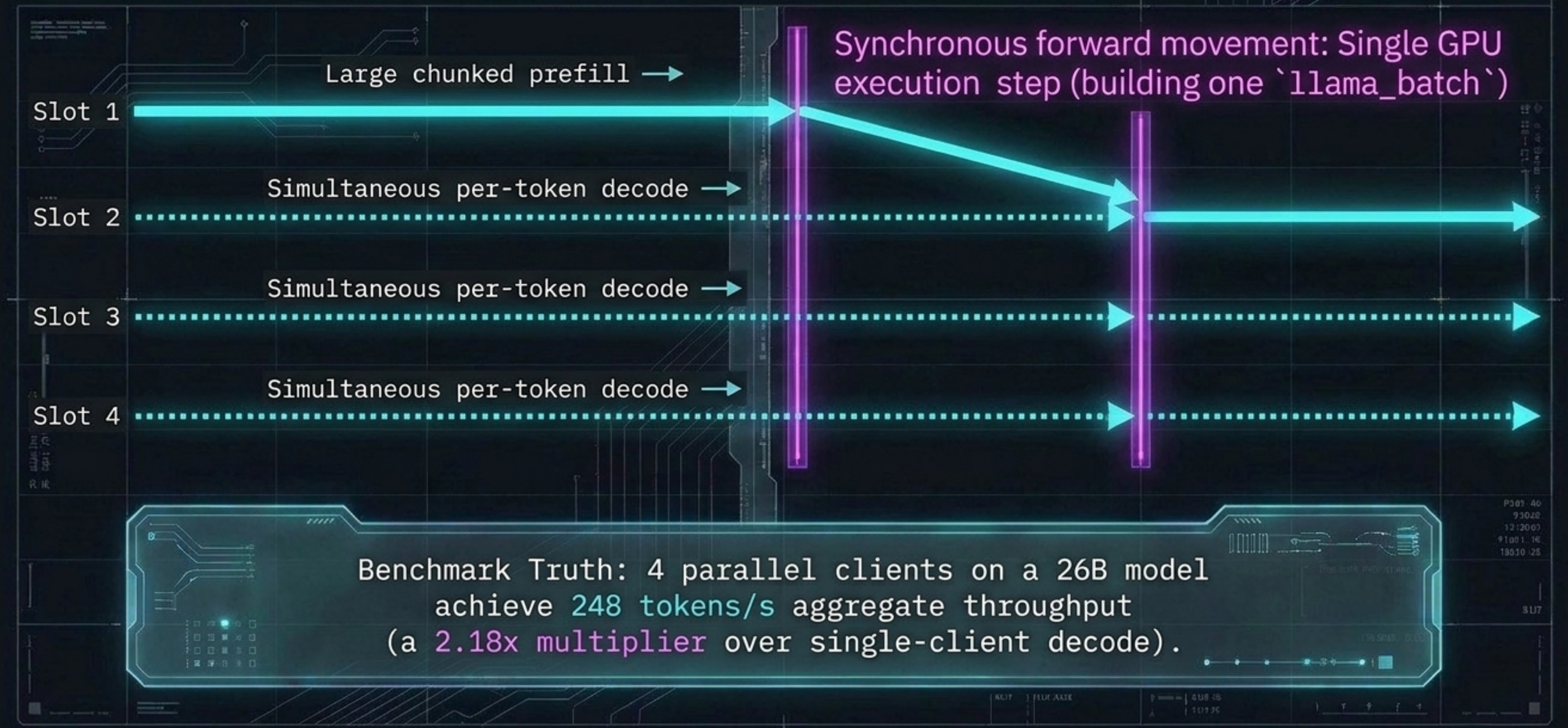




1. Moving away from sequential resource locking.
2. One persistent worker thread per resident model.
3. Context capacity pre-allocated by the LayerPlacementPlan via `n_seq_max = slots`.

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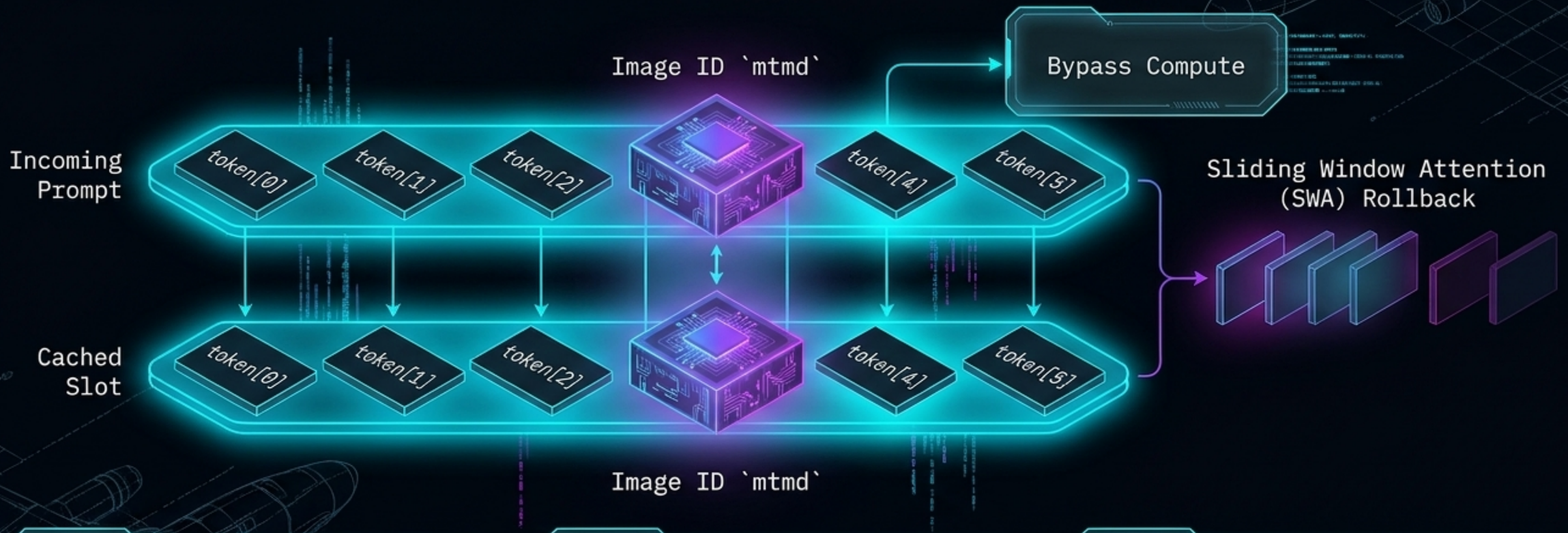
High-Throughput Continuous Batching Engine



Benchmark Truth: 4 parallel clients on a 26B model achieve 248 tokens/s aggregate throughput (a 2.18x multiplier over single-client decode).

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Multi-Modal Prefix Caching & Sequence Matching



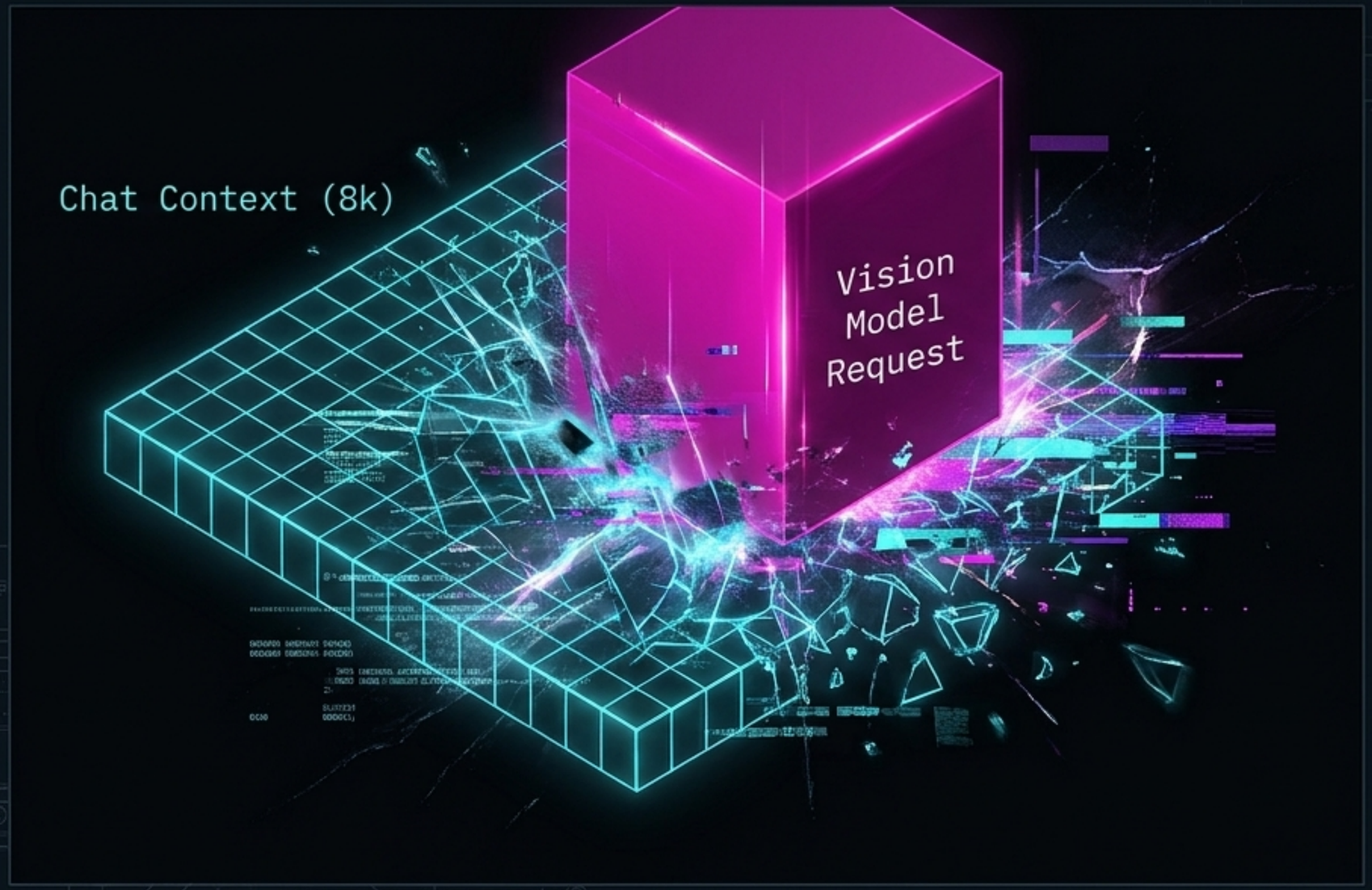
Matches text tokens AND multimodal chunk IDs.

An image already in the prefix is never re-encoded by the vision tower.

Benchmark Truth: TTFT drops from 306ms to 101ms for follow-up vision turns.

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The Eviction Constraint: Context Destruction



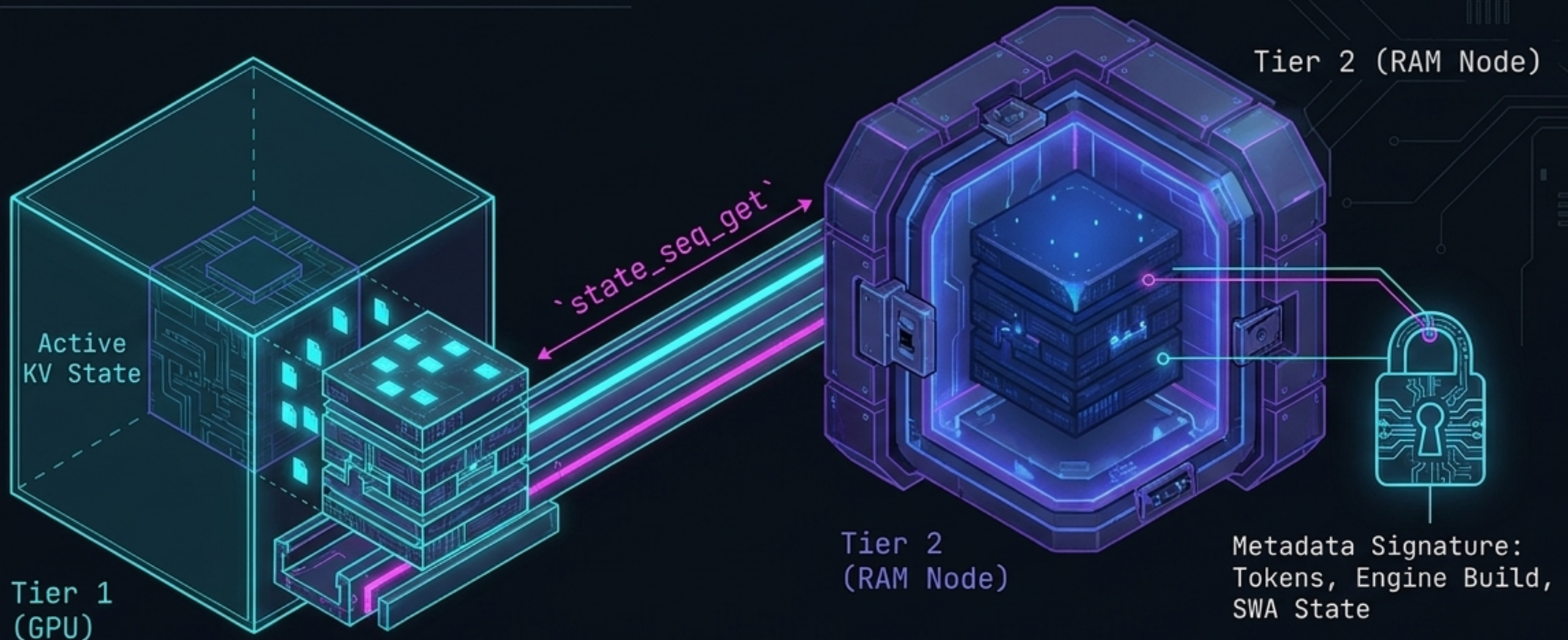
> ⚠ Warning
>
> Demotion traditionally destroys the worker thread, the context, and the KV cache.

> ⚠ Warning
>
> The Cost: The next chat turn pays a massive re-prefill penalty (e.g., 2.6 seconds of pure compute just to recover state).

NETWORK TELEMETRY
VRAM_EVICT: COMPLETE
STATE_LOST: 8192 TOKENS
COMPUTE_PENALTY: HIGH

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Architectural Breakthrough: Paginated KV State Parking

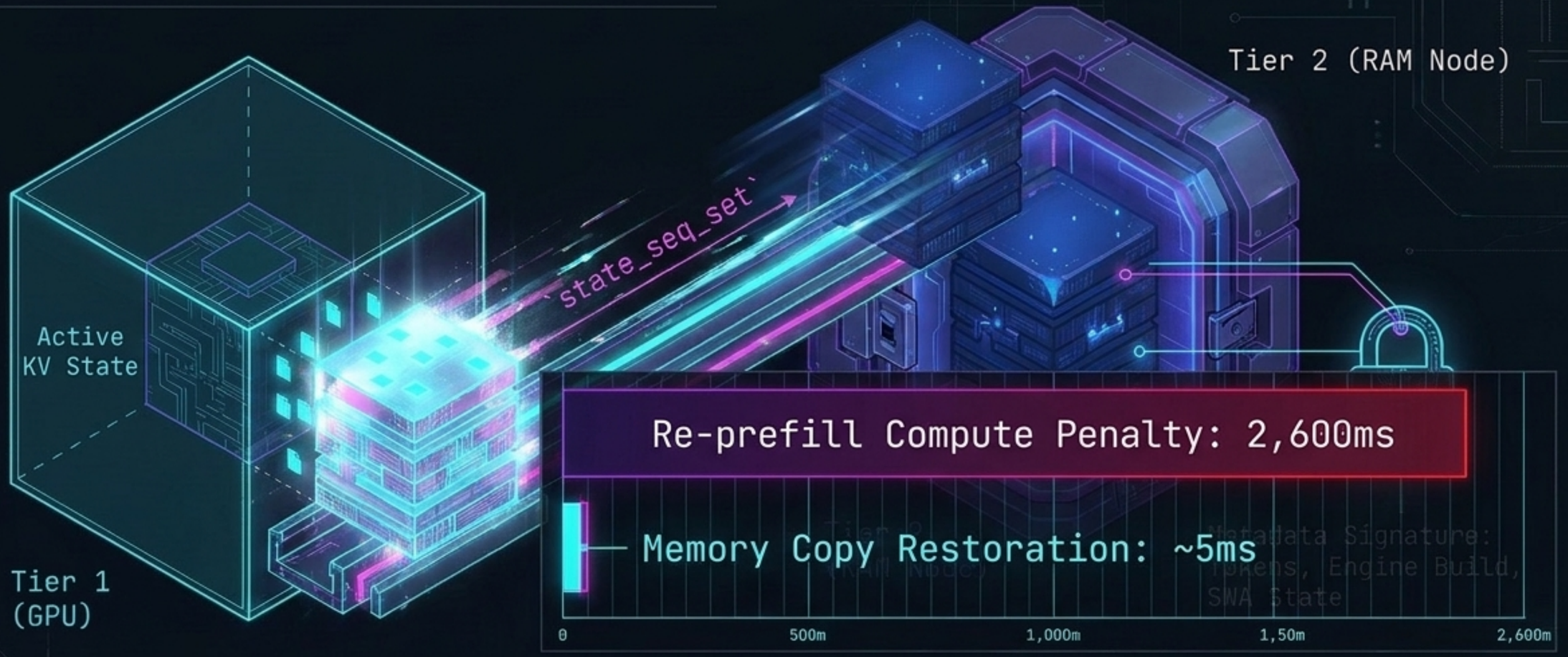


Preserving active conversational contexts across hardware tier switches.

Extraction footprint: ~360MB for an 8k context.

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Thawing the Cache: Near-Instant State Restoration



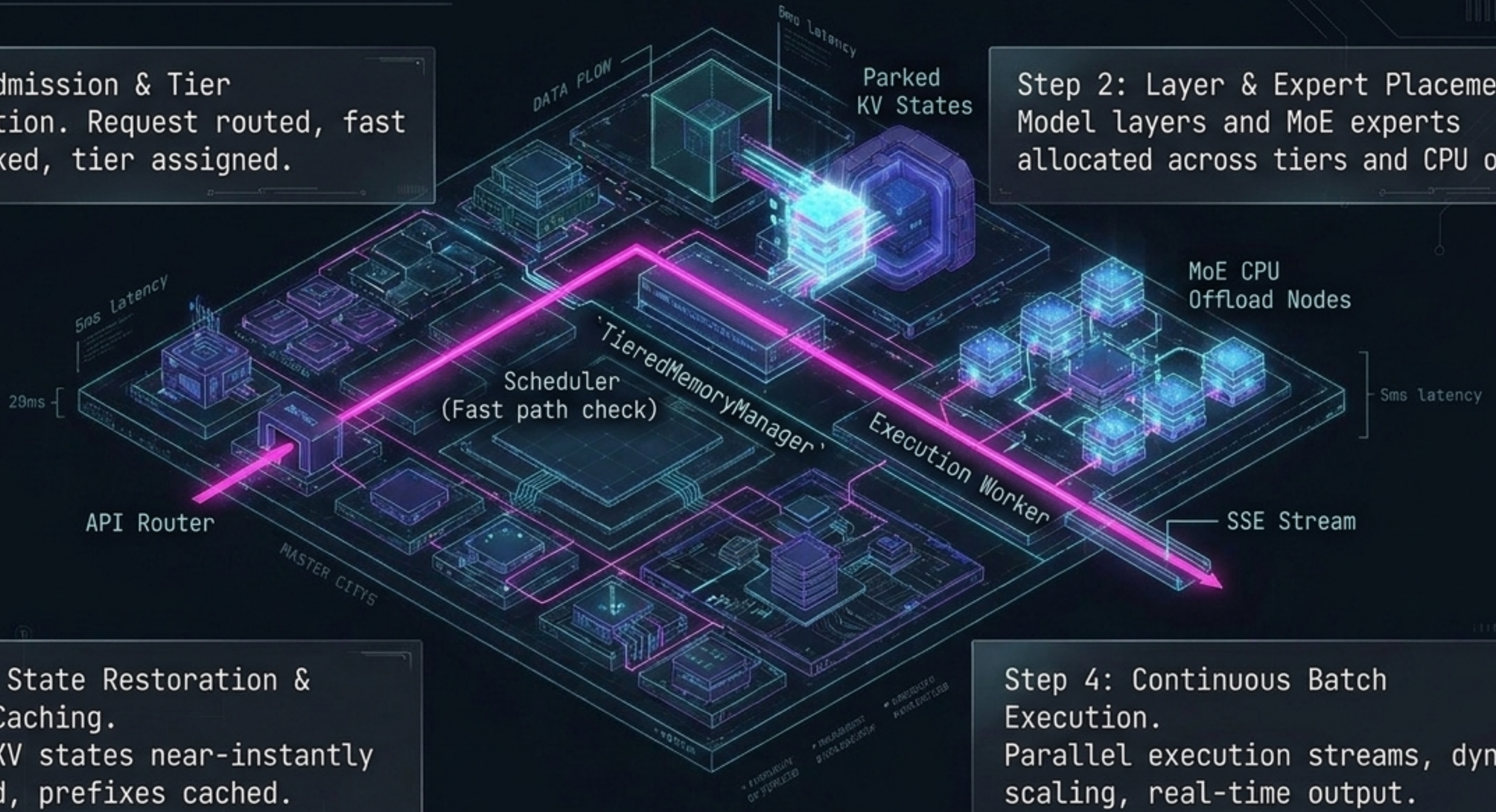
The Math: Restoring an 8k context requires a 360MB memory copy.
Benchmark Truth: TTFT for a swapped 12.6k context drops from 4.7 seconds down to 1.01 seconds.

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Synthesis:
The Request Lifecycle

Step 1: Admission & Tier Orchestration. Request routed, fast path checked, tier assigned.

Step 2: Layer & Expert Placement. Model layers and MoE experts allocated across tiers and CPU offload.

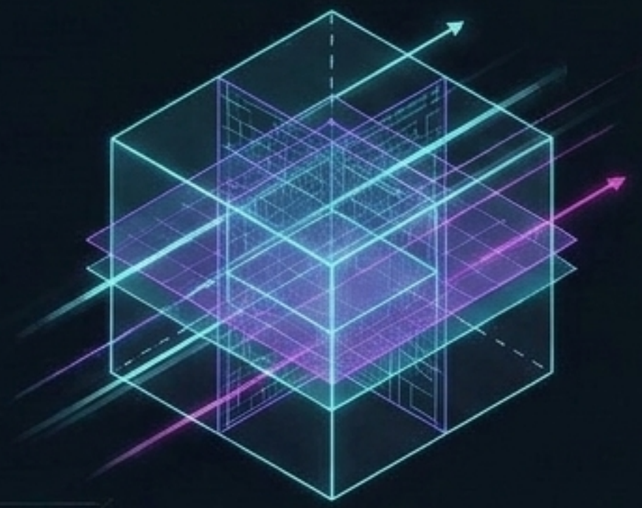


Step 3: State Restoration & Prefix Caching.
Parked KV states near-instantly restored, prefixes cached.

Step 4: Continuous Batch Execution.
Parallel execution streams, dynamic scaling, real-time output.

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The Mechanical Truth of Hypnos



Defeating the Memory Wall: Total hardware utilization across 3 tiers.



Eradicating Recompute: Persistent, parked execution contexts.



True Multi-Modal Density: In-process engine fusion with zero network overhead.

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